



# CREDIT CARD FRAUD DETECTION

Supervised by: Dr. Ahmed Magdy

محمد ثروت

ياسين مصطفى

مي محمد ابراهيم

منى جهاد انور

محمد عصام

منه محمد غريب

يوسف خويطر

محمد محمود

نادين احمد عبد النبي

# PROBLEM STATEMENT

## The Challenge

Financial fraud accounts for billions in losses annually. Traditional rule-based systems are static and easily bypassed by modern criminal tactics.

**Objective:** To develop an adaptive machine learning model that analyzes behavioral patterns to predict and prevent fraudulent transactions in real-time.



# DATASET OVERVIEW



## 7,000 Records

A comprehensive collection of credit card transaction instances.



## 12 Attributes

Features include Age, Distance, Time, and Device Type.



## High Bias

Imbalanced distribution mimicking real-world rarity of fraud.

# DATA PREPROCESSING

## Cleaning & Formatting

- ✓ **Missing Values:** Handled using `ReplaceMissingValues` to account for 5% null data.
- ✓ **Type Conversion:** Applied `NumericToNominal` to correctly categorize the target class.
- ✓ **Consistency:** Ensuring all features are normalized for algorithm consumption.



# THE INITIAL "ACCURACY TRAP"

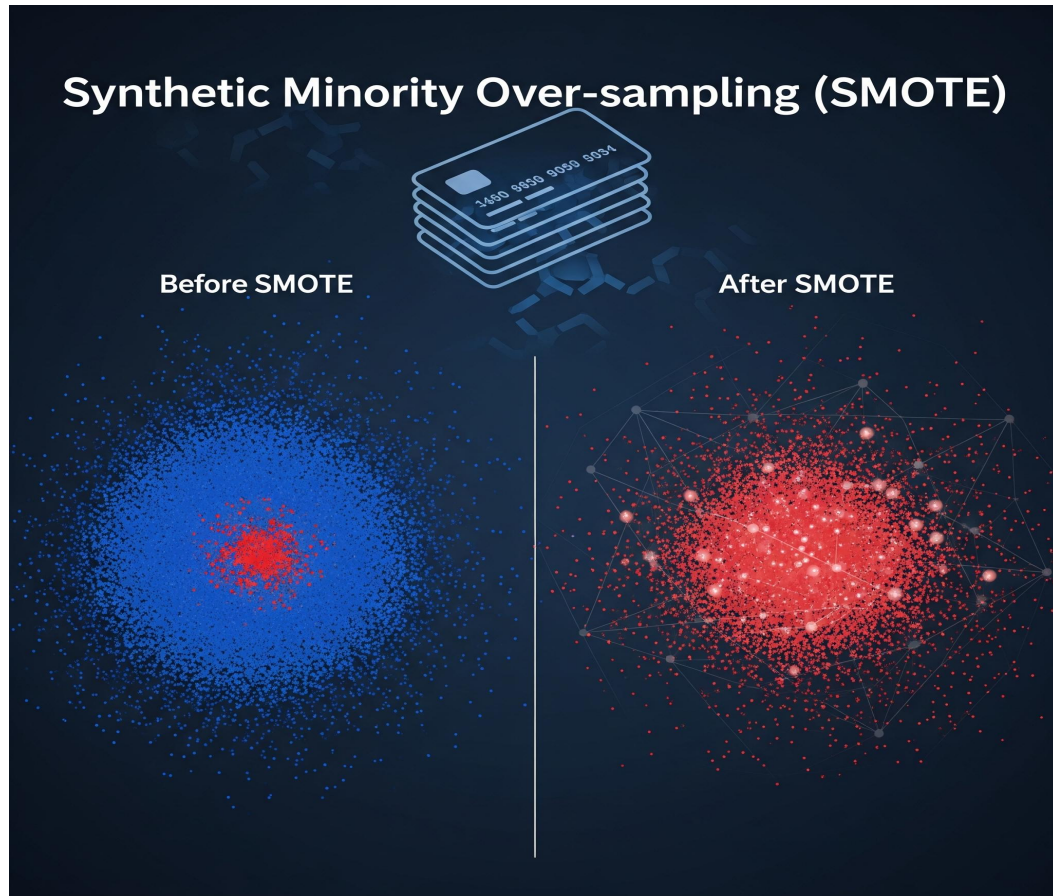
## 89.7% Accuracy Score

At first glance, the model appeared highly successful. However, the Confusion Matrix revealed a critical flaw.

The model detected **0 Fraud Cases**. It simply learned to guess "Genuine" for every instance to achieve a high score, ignoring the minority class entirely.

| Predicted | Genuine | Fraud |
|-----------|---------|-------|
| Genuine   | 6,279   | 0     |
| Fraud     | 721     | 0     |

# SOLUTION: SMOTE BALANCING



## Synthetic Minority Over-sampling

We applied the **SMOTE** algorithm to generate synthetic fraud instances based on existing feature patterns.

- Multiplied minority instances by **800%**.
- Created a balanced environment for the classifier.
- Forced the model to learn fraud "signatures" rather than simple probability.

# ALGORITHM: J48 DECISION TREE



## Entropy

Minimizing disorder to find the most informative features.



## Pruning

Removing weak branches to prevent overfitting and ensure speed.



## Rules

Generated 203 logical rules across 405 nodes.

# PERFORMANCE METRICS

Accuracy (Total Precision)

**90.77 %**

(TP+TN) / Total

Precision (Integrity)

**95.96 %**

TP / (TP+FP)

Recall (Sensitivity)

**85.43 %**

TP / (TP+FN)

Specificity

**96.28 %**

TN / (TN+FP)

TP: 5544 | TN: 6846 | FP: 233 | FN: 945

=== Summary ===

|                                  |           |           |
|----------------------------------|-----------|-----------|
| Correctly Classified Instances   | 11590     | 90.7738 % |
| Incorrectly Classified Instances | 1178      | 9.2262 %  |
| Kappa statistic                  | 0.8158    |           |
| Mean absolute error              | 0.1311    |           |
| Root mean squared error          | 0.2836    |           |
| Relative absolute error          | 26.2233 % |           |
| Root relative squared error      | 56.7352 % |           |
| Total Number of Instances        | 12768     |           |

=== Detailed Accuracy By Class ===

|               | TP Rate | FP Rate | Precision | Recall | F-Measure | MCC   | ROC Area | PRC Area | Class |
|---------------|---------|---------|-----------|--------|-----------|-------|----------|----------|-------|
|               | 0.963   | 0.146   | 0.865     | 0.963  | 0.911     | 0.821 | 0.923    | 0.867    | 0     |
|               | 0.854   | 0.037   | 0.960     | 0.854  | 0.904     | 0.821 | 0.923    | 0.938    | 1     |
| Weighted Avg. | 0.908   | 0.090   | 0.913     | 0.908  | 0.908     | 0.821 | 0.923    | 0.903    |       |

=== Confusion Matrix ===

```
a    b  <-- classified as
6046 233 | a = 0
945 5544 | b = 1
```



# IMPACT OF DATA ENGINEERING



*The optimized system correctly identified 5,544 fraudulent transactions previously missed by the baseline model.*

# PREDICTIVE INSIGHTS

## Key Indicators of Fraud

The Decision Tree identified high-risk correlations:

- **Distance Factor:** Transactions far from the billing address increased risk scores significantly.
- **Velocity Score:** Frequent high-value transactions in short periods were key indicators.
- **Device History:** New or unrecognized devices triggered higher security flags.

### Logic Sample:

```
IF (distance > 18.2) AND  
(velocity > 6.69) AND  
(first_transaction = TRUE)  
THEN is_fraud = 1
```

# CONCLUSION

We successfully transformed a biased and failing model into a highly precise financial shield.

95.9%

Precision

85.4%

Recall

90.7%

Accuracy

This project demonstrates that in financial AI, **Data Balancing (SMOTE)** is as critical as the algorithm selection itself.

# THANK YOU!

Questions & Discussion

---

# DATA SOURCES & REPOSITORY

## Kaggle Dataset

Credit Card Fraud Detection Dataset by Waddah Ali  
<https://www.kaggle.com/datasets/waddahali/fraud-detection>

## GitHub Repository

Project Implementation & Codebase  
[\[Insert GitHub Project Link Here\]](#)

## Research Context

Information derived from financial AI fraud detection methodologies.